be recorded by detector 4, but if it heads along path C it will be detected by either detector 3 or detector 2, depending on the path it follows after passing through beam splitter c.

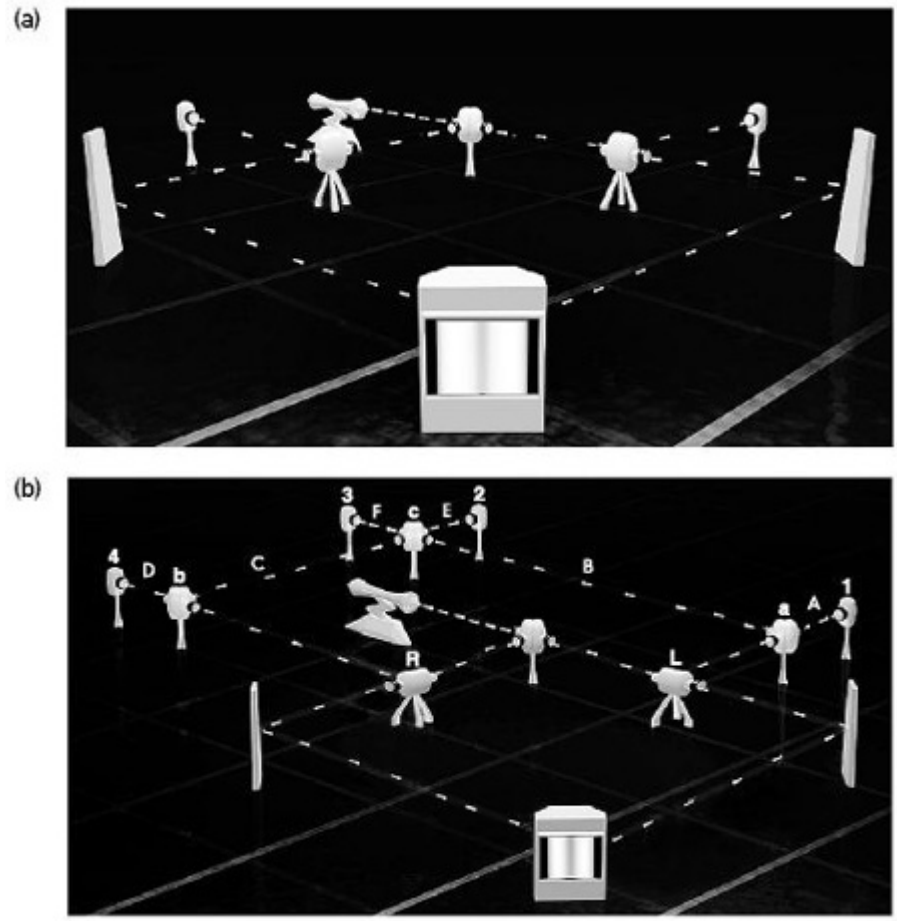


Figure 7.5 (a) A beam-splitter experiment, augmented by down-converters, does not yield an interference pattern, since the idler photons yield which-path information. (b) If the idler photons are not detected directly, but instead are sent through the maze depicted, then an interference pattern can be extracted from the data. Idler photons that are detected by detectors 2 or 3 do not yield which-path information and hence their signal photons fill out an interference pattern.

Now for why we've added all this complication. Notice that if an idler photon is detected by detector 1, we learn that the corresponding signal photon took the left path, since there is no way for an idler that was emitted from down-converter R to find its way to this detector. Similarly, if an idler photon is detected by detector 4, we learn that its signal photon partner took the right path. But if an idler photon winds up in